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# Circulating extracellular vesicles-microRNAs as potential biomarkers for the identification of ME/CFS: differentiating fatigue-related conditions

Akiko Eguchi<sup>1,2\*</sup> , Hirohiko Kuratsune<sup>3</sup>, Yasuhito Nakatomi<sup>4,5</sup>, Takao Yasui<sup>6</sup>, Ryo Nakagawa<sup>7</sup>, Yasuyoshi Watanabe<sup>8,9</sup> and Sanae Fukuda<sup>10</sup>

## Abstract

**Background** Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a debilitating, multi-systemic condition that has gained renewed global attention due to its clinical overlap with the growing population of long COVID patients. Despite ongoing efforts to characterize the disease, definitive diagnostic molecular markers are yet to be fully established, posing challenges in clinically differentiating from idiopathic chronic fatigue (ICF) and depression (Dep). This study aimed to identify circulating extracellular vesicles (EVs)-associated microRNAs (miRNAs) that serve as both diagnostic signatures and windows into the disease's underlying pathophysiology.

**Methods** Circulating EVs from ME/CFS ( $n=6$ ), ICF ( $n=6$ ), and depression ( $n=8$ ) patients were analyzed using flow cytometry, nano-tracking analysis, and comprehensive miRNA analysis. Differentially expressed miRNAs were analyzed using KEGG pathway enrichment to identify ME/CFS-specific signatures. Key candidate biomarkers were further validated in an additional healthy control (HC) cohort ( $n=4$ ).

**Results** ME/CFS-EVs exhibited a unique subpopulation with high calcein intensity and larger diameters. Initial global miRNA profiling (Volcano plot) identified miR-21-5p, let-7f-5p, miR-26b-5p, and miR-20a-5p as significantly dysregulated EV-miRNAs in ME/CFS compared to ICF and Dep. To explore systemic pathophysiology, we identified a 114 EV-miRNA signature that achieved  $87.0 \pm 4.8\%$  sensitivity and  $93.7 \pm 2.4\%$  specificity within repeated cross-validation of the discovery cohort. After adjusting for covariates, 91 miRNAs remained significant; pathway analysis of the 62 up-regulated EV-miRNAs revealed significant enrichment in neuro-systemic axes, encompassing cellular structural integrity (focal adhesion), core signaling hubs (PI3K-Akt), and systemic homeostasis (such as insulin signaling and endocrine functions). Preliminary evaluation confirmed that these target EV-miRNAs remained at minimal or undetectable levels in the HC group.

**Conclusions** A 62 EV-miRNA signature provides insight into the interconnected neuro-systemic pathways disrupted in ME/CFS, particularly those governing neuronal connectivity and cellular scaffolding. Within this candidate

\*Correspondence:  
Akiko Eguchi  
akieguchi@med.mie-u.ac.jp

Full list of author information is available at the end of the article



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EV-miRNA signature, the top-ranked miRNAs—miR-21-5p, let-7f-5p, miR-26b-5p, and miR-20a-5p—emerge as potential candidate biomarkers whose specific elevation was not shared by HC. These findings establish a valuable framework for targeted diagnosis and enhance our understanding of the molecular pathways involved in synaptic and structural alterations in ME/CFS.

**Keywords** ME/CFS, Circulating EV, Non-invasive biomarkers, Abnormal neuronal pathway

## Background

Myalgic encephalomyelitis/chronic fatigue syndrome (ME/CFS) is a serious and complex debilitating disease characterized by a wide spectrum of symptoms, including post-exertional malaise (PEM), chronic pain, and neurocognitive deterioration [1–4]. Recent comprehensive reviews emphasize that ME/CFS is a multi-factorial complex disease involving interplay between genetics, environment, immunity, metabolism, and gut microbiota [5], although diagnosis remains primarily symptom-based due to the absence of definitive biomarkers [4]. The diagnosis of ME/CFS relies on clinical symptoms spanning a broad range of severity. Large-scale evidence mapping of over 610,000 cases highlights significant inconsistencies across studies due to diverse diagnostic criteria, underscoring the urgent need for standardized objective markers [6]. Achieving a definitive and non-invasive diagnosis remains difficult for clinicians due to the reliance on subjective symptoms and the lack of established laboratory tests. While several biological abnormalities have been recognized as important contributors – including impaired calcium ion channels [7], dysregulation of the neuroimmune system [8], glial cell activation [9], and metabolic/mitochondrial dysfunction [10, 11]– the precise pathogenic mechanisms are not yet fully understood.

Recent clinical observations from the COVID-19 pandemic have underscored the importance of identifying ME/CFS biomarkers, as an estimated 26 to 50% of patients in clinic-based long COVID (post COVID condition) cohorts meet the diagnostic criteria for ME/CFS [12–14]. Although various potential biomarkers have been proposed [15] – including autonomic dysfunction, circulating cytokines, cellular factors/signalings [16], reactivation of herpesviruses such as Epstein-Barr virus [17], and oxidative stress markers – additional objective markers are required to distinguish ME/CFS from other fatigue-related conditions, including idiopathic chronic fatigue (ICF) and clinical depression. Furthermore, as international clinical guidelines have recently been updated to advise a more tailored approach to graded exercise therapy [18], and recent studies suggest that the suitability of such therapies can vary significantly among patients [4], there is a critical, long-awaited need for a breakthrough in understanding ME/CFS etiology and identifying objective biomarkers to guide personalized treatment [19]. Significant momentum has been gained from deep phenotyping studies, such as the NIH

large-scale intramural study, which provides detailed physiological and immunological insights into post-infectious ME/CFS, helping to address the long-standing “legitimacy deficit” of the disease [20].

Extracellular vesicles (EVs) are released from damaged or stressed cells and circulate in the bloodstream, carrying cellular cargo such as proteins, lipids, and miRNAs [21]. EVs are recognized as stable, non-invasive biomarkers for various diseases because they reflect the physiological state of their cells of origin. Furthermore, EV function as mediators in cell-to-cell and organ-to-organ communication by delivering their molecular content to target cells, thereby modulating systemic signaling pathways [22, 23]. We previously reported that circulating EV-proteins associated with the actin network – including talin-1, filamin-A, and 14-3-3 family proteins – are highly specific biomarkers that can distinguish ME/CFS from ICF and depression [24]. These findings suggested a systemic modulation of cellular integrity; however, the regulatory mechanisms underlying these protein changes and their link to neuroinflammation remained unclear. Emerging evidence indicates that specific miRNAs, particularly the let-7 family, can act as endogenous ligands for Toll-like receptor 7 (TLR7) [25], directly triggering microglia-mediated neuroinflammation previously observed in ME/CFS patients via PET imaging [9]. Furthermore, dysregulation of the PI3K-Akt signaling pathway from EV-protein comprehensive analysis has been implicated in the metabolic and immune dysfunction of ME/CFS [24].

In this study, we performed a comprehensive small RNA sequencing of circulating EVs to identify ME/CFS-specific miRNA signatures. By comparing ME/CFS with both ICF and depression, we aimed to determine whether EV-associated miRNAs reflect unique pathophysiological processes, such as deeper understanding of the disease etiology.

## Methods

### Subjects and study design

The study was approved by the ethics committees of Kansai University Welfare of Health Sciences (Approval No. 09–06), Osaka City University Graduate School of Medicine (Approval No. 2151), and Mie University (Approval No. 1697 and H2018-069), and was conducted in accordance with the Declaration of Helsinki. All subjects, including ME/CFS patients ( $n=6$ ), ICF patients

( $n=6$ ), and depression patients ( $n=8$ ), provided written informed consent before enrolment. ME/CFS, ICF, and depression patients who visited the outpatient clinic of Osaka City University Hospitals and Clinic B were randomly enrolled into the study. Healthy individuals ( $n=4$ ) who visited the outpatient Health Checkup Center were randomly enrolled into the study. ME/CFS and ICF were diagnosed based on the 1994 Centers for Disease Control and Prevention (CDC) clinical criteria (Fukuda criteria) [26] and the Canadian Consensus Case Definition [27] by specialists at the Osaka City University Hospitals and Clinic B. Depression was assessed using the Structured Clinical Interview for DSM-IV. Exclusion criteria were as follows: (1) neuro-inflammatory or immune disorders diagnosed by clinical laboratory tests and magnetic resonance imaging; (2) any active medical condition that could explain the presence of chronic fatigue; (3) presence of any diagnosable illness that relapsed or was not completely resolved, such as certain types of malignancy or chronic cases of hepatitis B or C virus infection; (4) alcohol or other substance abuse; (5) severe obesity (body mass index  $\geq 30$  kg/m<sup>2</sup>); (6) pregnancy; or (7) lactation. The presence of major depressive disorder, fibromyalgia, or somatoform disorder was not a criterion for exclusion. Abdominal discomfort syndrome was not assessed. Psychiatric disorders associated with CFS symptoms were diagnosed by psychiatrists and blood samples were collected at Osaka City University Hospital and Clinic B.

#### Measuring circulating EV number using flow cytometry analysis

Circulating EV numbers were determined as previously described [28]. Briefly, sample was first centrifuged at  $2000 \times g$  for 10 min to eliminate any aggregation. After centrifugation, circulating EVs were stained with 4  $\mu$ g/ml of calcein-AM solution (Invitrogen, San Diego, CA) in the dark at room temperature. Calcein-AM is internalized and hydrolyzed to a fluorescent product by intravesicular esterases, allowing for the detection of intact EVs. The number of EVs was determined via flow cytometry (BD Canto II; BD Biosciences, San Jose, CA) calibrated with ultraviolet 2.5  $\mu$ m Alignflow alignment beads (Thermo Fisher Scientific, Tokyo, Japan) in triplicate for each sample, and data were analyzed using FlowJo software (TreeStar, Ashland, OR).

#### EV characterization ~ EV-miRNA analysis

To isolate EVs, sample was centrifuged at  $2000 \times g$  for 10 min to eliminate any aggregations. Circulating EVs were then isolated via qEV columns (Izon Science, Christchurch, New Zealand) according to the manufacturer's instructions. Thirty fractions (500  $\mu$ l each) were collected, and the protein concentration was monitored by measuring OD<sub>280</sub>. Fractions 6–10, which

corresponded to the first peak of OD<sub>280</sub> and were defined as EV-rich fractions (by the manufacturer's instruction), were concentrated using Amicon Ultracel-3 K (EMD Millipore, Temecula, CA). The size distribution of isolated EVs was measured in triplicate by nanoparticle tracking analysis (NTA) using Nanosight LA10 (Malvern Panalytical, UK).

Total RNA from the isolated circulating EVs was extracted using the miRNeasy kit (Qiagen, Japan) and analyzed using an Agilent bioanalyzer/electrophoresis at TORAY (Tokyo, Japan). Comprehensive miRNA profiling was performed using the TORAY 3D-Gene human miRNA Oligo Chip. Raw data normalization and analysis were performed at TORAY. Briefly, for miRNA expression analysis, raw signal intensities were processed and normalized using GeneSpring GX software (Agilent Technologies, Santa Clara, CA, USA). The data were normalized using the 75th percentile normalization method, where the 75th percentile value of the signal intensities was set to 1, and subsequently converted to Log2 values. For differential expression analysis, a  $p$ -value was calculated ( $p = |\text{Log2 fold change}| \times -\text{Log10}(p\text{-value})$ ) to identify significant miRNAs. A heat map was generated using Heatmapper 2 [29]. KEGG pathway analysis was performed using the DIANA-miRPath software [30].

#### Machine learning-based classification of EV-miRNA profiles

The classification of disease groups based on EV-miRNA expression profiles was performed using a custom-developed application implementing a supervised machine learning framework. Normalized miRNA expression values were used as input features. Feature selection was first performed by differential expression analysis using adjusted  $p < 0.001$ , yielding 114 candidate miRNAs for the primary analysis. A secondary sensitivity analysis was conducted using 550 candidate miRNAs selected with adjusted  $p < 0.05$ . This analytical pipeline is consistent with previously established approaches for multi-miRNA-based classification [31], enabling systematic integration of high-dimensional miRNA data for disease discrimination. Thus, the framework provides a structured approach for identifying disease-associated miRNA patterns.

A logistic regression-based classifier with L1 regularization was employed to construct the diagnostic model. Within the predefined candidate miRNA set, L1 regularization further selected a compact subset of informative features by shrinking non-informative coefficients to zero. The regularization parameter ( $\lambda = 1.0$ ) and the coefficient-selection threshold (0.1) were fixed throughout the analysis. Only miRNA species with nonzero coefficients after regularization were retained as explanatory variables for classification.

Model performance was evaluated using repeated three-fold cross-validation (nine repetitions). During each cross-validation iteration, coefficient estimation and model fitting were performed exclusively on the training subset, and the resulting model was applied to the corresponding validation subset. The procedure was repeated across all folds and repetitions to reduce variability arising from different data partitions and to evaluate the stability of feature selection. Because the initial differential-expression filtering step was performed before cross-validation, the resulting performance estimates should be interpreted as exploratory and may contain some degree of optimistic bias.

Performance metrics included sensitivity (recall), specificity, precision, accuracy, ROC curves, class-specific AUC values, and aggregated confusion matrices calculated from prediction results obtained across all repeated cross-validation runs. The proposed framework should therefore be regarded as an exploratory machine-learning approach for identifying candidate EV-miRNA signatures, and independent validation will be required before clinical applicability can be established.

Statistics

All data are expressed as mean ± standard error of the mean (SEM) unless otherwise indicated. Statistical significance among the three groups (ME/CFS, ICF, and depression patients) was determined using ANOVA, followed by Tukey’s post-hoc test for pairwise comparisons (ME/CFS vs. ICF and ME/CFS vs. depression). To control for the false discovery rate (FDR) in multiple-testing, P-values were adjusted using the Benjamini-Hochberg method. To account for potential confounding effects, an analysis of covariance (ANCOVA) was performed with age, sex, and disease duration as covariates. To determine the magnitude of the differences between the groups, effect sizes were calculated using Hedges’ g. All statistical analyses were performed using GraphPad Prism (version 10, GraphPad Software, San Diego, CA, USA). An adjusted p-value < 0.05 was considered statistically significant.

**Table 1** Patient demographics in ME/CFS, ICF, and depression

| Variables                                 | ME/CFS (n=6)                           | ICF (n=6)    | Dep (n=8)    |
|-------------------------------------------|----------------------------------------|--------------|--------------|
| Gender (male/female)                      | 0/6                                    | 4/2          | 2/6          |
| Age (years)                               | 40.33 ± 6.12                           | 38.67 ± 7.87 | 40.63 ± 10.8 |
| Approximately Duration of disease (years) | 14.20 ± 5.81                           | 7.38 ± 5.46  | 0.50 ± 0.18  |
| Performance status (PS 0–9)               | PS 4 (n=1)<br>PS 6 (n=3)<br>PS 8 (n=2) | N/A          | N/A          |

Abbreviations: ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome; ICF, idiopathic chronic fatigue; Dep, depression; N/A, not applicable

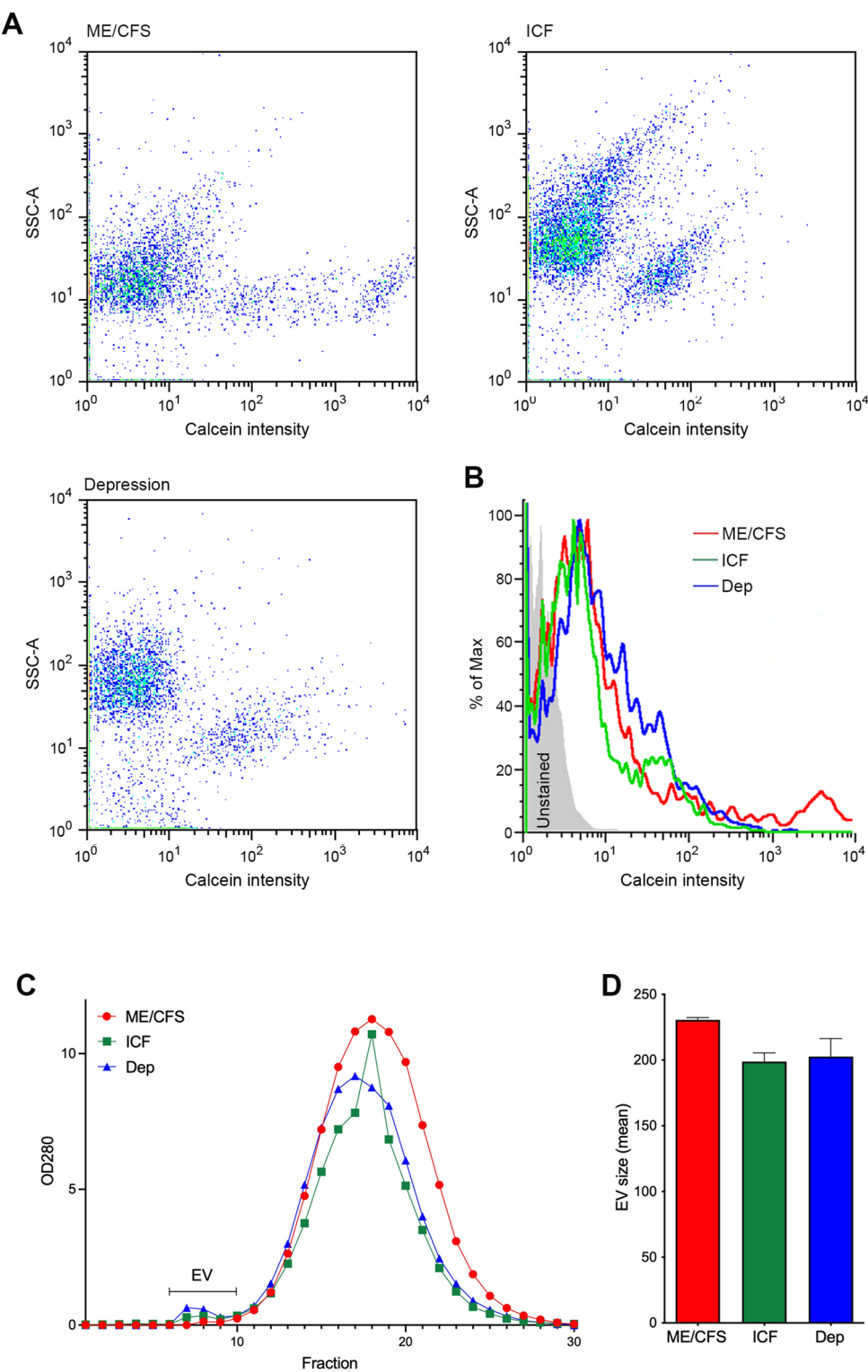
Results

Clinical characteristics of the study cohorts

Our previous findings demonstrated that ME/CFS patients exhibit a significantly higher number of circulating EVs compared to healthy individuals, providing a diagnostic accuracy of 90–94% based on the area under the curve (AUC) [24]. Additionally, we identified EV-proteins related to actin-network proteins, including talin-1 and filamin-A, as highly specific ME/CFS biomarkers. Although the EV-protein profile indicated cellular damage, there was insufficient evidence to identify the specific tissues or pathways involved in ME/CFS progression. These results led us to investigate whether circulating EV-microRNAs (miRNAs) could distinguish ME/CFS from ICF and depression (Dep) - two conditions also associated with fatigue. Patient demographics are summarized in Table 1. All (100%) ME/CFS patients were female, whereas 66% of ICF and 33% of depression patients were male; this aligns with the clinical profile of ME/CFS, where approximately 80% of patients are female. The average age was comparable across groups (ME/CFS: 40.33 ± 6.12 years; ICF: 38.67 ± 7.87; depression: 40.63 ± 10.8). The approximate duration of disease in ME/CFS (14.20 ± 5.81 years) was longer than in ICF (7.38 ± 5.46 years) and depression (0.50 ± 0.18 years), reflecting the chronic nature and the lack of effective treatments for ME/CFS. The severity of ME/CFS was assessed by Performance Status (PS 0–9), with patient distributed as follows: one individual was at PS 4 (capable of light work for several days/week), three were at PS 6 (requiring home rest > 50% of the week and frequent assistance), and two were at PS 8 (bedbound for most of the day and requiring constant assistance).

Differential physical and enzymatic properties of circulating EVs in ME/CFS

To characterize EVs in peripheral blood, we performed flow cytometric analysis using Calcein staining, which identifies enzymatically active and intact EVs [28]. EVs were gated based on side scatter (SSC-A) and Calcein fluorescence intensity. The ME/CFS group exhibited a distinct distribution profile compared to the ICF and Dep groups (Fig. 1A and B). While all groups displayed EV populations with low and moderate Calcein intensity, a unique and prominent third peak characterized by high Calcein intensity (approximately 10<sup>3</sup> – 10<sup>4</sup>) was specifically observed in the ME/CFS samples (Fig. 1A and B). In contrast, this high-intensity signal was minimal or absent in the ICF and Dep groups (Fig. 1A and B). For further characterization, circulating EVs were purified from six ME/CFS, six ICF, and eight Dep patients using qEV size-exclusion chromatography columns to remove contaminating free proteins (representative data shown in Fig. 1C). The diameter of the circulating EVs was around



**Fig. 1** Physical and enzymatic characterization of circulating extracellular vesicles (EVs). **(A)** Dot plots of SSC-A versus Calcein intensity. **(B)** Representative flow cytometry histograms of circulating EVs stained with Calcein-AM. The ME/CFS group shows a unique third peak with high Calcein intensity (red arrow) compared to ICF and depression (Dep) groups. **(C)** Representative qEV size-exclusion chromatography profile. The first OD280 peak (Fractions 6–10) corresponds to the EV-rich fractions used for subsequent analyses. **(D)** Size distribution and concentration of isolated EVs measured by nanoparticle tracking analysis (NTA). ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome; ICF, idiopathic chronic fatigue; Dep, depression; EVs, extracellular vesicles

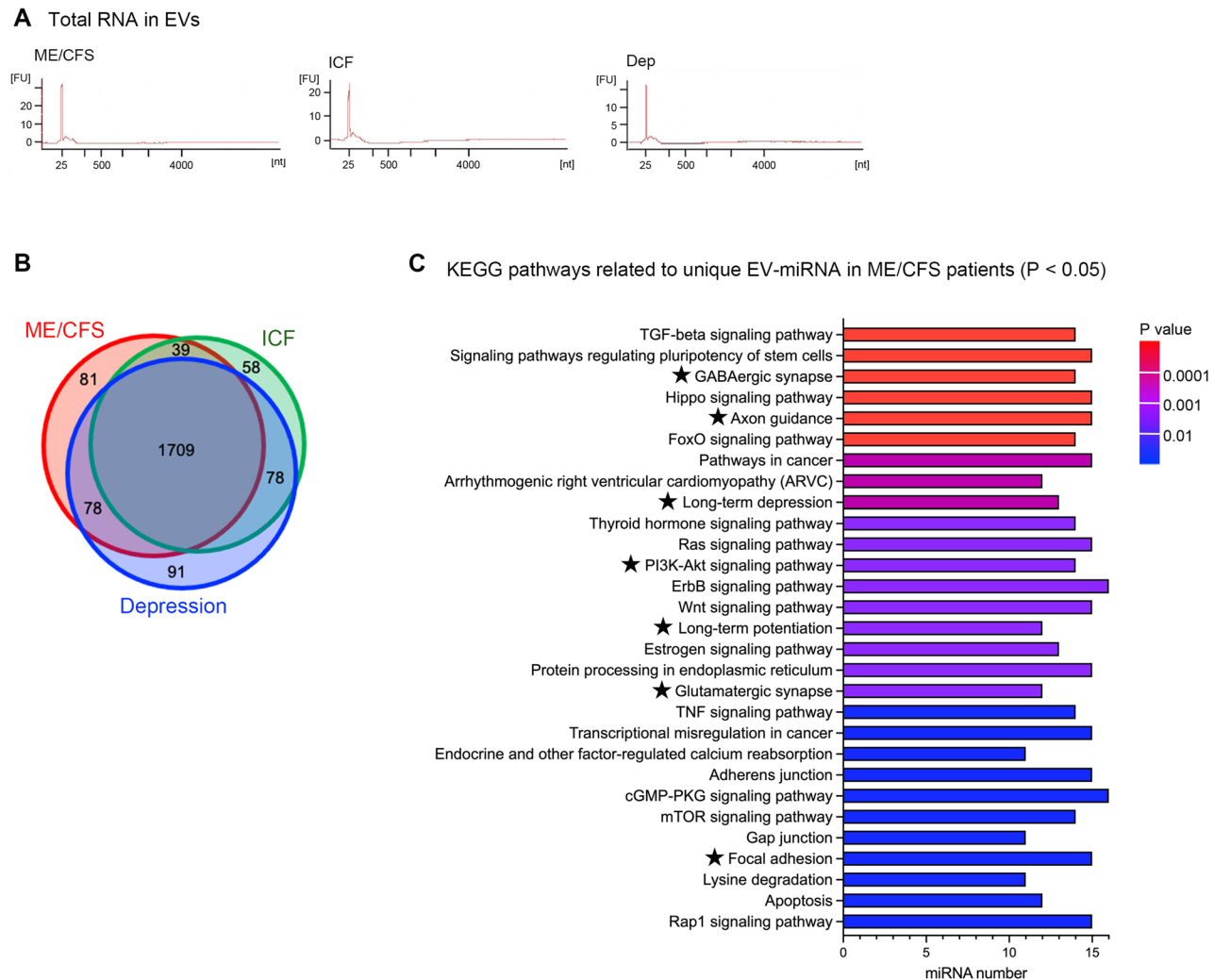


200 nm across all groups (Fig. 1D). However, EVs from the ME/CFS group were larger (mean: 230 nm) compared to those from the ICF (199 nm) and Depression (202 nm) groups, as determined by nano tracking analysis (Fig. 1D). These results indicate that the high-intensity Calcein-positive EV subpopulation is a specific feature of ME/CFS, suggesting that these EVs may carry more cellular complex or dense molecular cargo.

Comprehensive analysis of miRNAs in circulating EVs

Given that uniquely high calcein-intensity and larger diameter were observed in ME/CFS EVs, we hypothesized that these EVs encapsulate a distinct set of molecular signatures. To test this, we focused on EV-miRNAs. Total RNA was isolated and characterized using an RNA analyzer. The representative profiles confirmed that the majority of isolated RNA consisted of small RNAs,

primarily miRNAs (Fig. 2A and Supplementary Fig. 1A). Comprehensive miRNA profiling identified 1,709 miRNAs common to all three groups (Fig. 2B). Additionally, 39 miRNAs were shared between the ME/CFS and ICF groups, 78 between the ME/CFS and depression groups, and 78 between the ICF and depression groups (Fig. 2B). To identify the qualitative differences between these conditions, we also focused on miRNAs that were exclusively detected in each group (Fig. 2B). Specifically, 81, 58, and 91 miRNAs were unique to the ME/CFS, ICF, and depression groups, respectively (Fig. 2B and Supplementary Table 1). KEGG pathway analysis revealed that these ME/CFS-unique miRNAs were involved in GABAergic synapse, axon guidance, long-term depression, long-term potentiation, and glutamatergic synapse (Fig. 2C and Supplementary Table 2). Notably, they were also linked to focal adhesion and the PI3K-Akt signaling pathway



**Fig. 2** Comprehensive miRNA profiling of circulating EVs. **(A)** Representative RNA profiles of EVs. **(B)** Venn diagram illustrating the number of common and unique miRNAs detected in the ME/CFS, ICF, and Dep groups. **(C)** KEGG pathway analysis of miRNAs exclusively detected in the ME/CFS group. Asterisks (★) denote pathways associated with neurological functions and those previously identified in our EV-proteomic analysis. ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome; ICF, idiopathic chronic fatigue; Dep, depression; EVs, extracellular vesicles; miRNA, microRNA

– consistent with the EV-protein signatures previously identified in ME/CFS (Fig. 2C). In contrast, unique miRNAs in the ICF group were primarily involved in long-term potentiation and glutamatergic synapse (Supplementary Tables 2 and Supplementary Fig. 1B), while those in the Dep group were linked to circadian entrainment, glutamatergic synapse, axon guidance, GABAergic synapse, and long-term depression (Supplementary Tables 2 and Supplementary Fig. 1C). These findings suggest that unique miRNAs in ME/CFS are linked to neuronal dysfunction and focal adhesion. This is consistent with our previous study, which revealed microglial activation via positron emission tomography (PET) and identified focal adhesion – related EV-proteins as diagnostic signatures for ME/CFS.

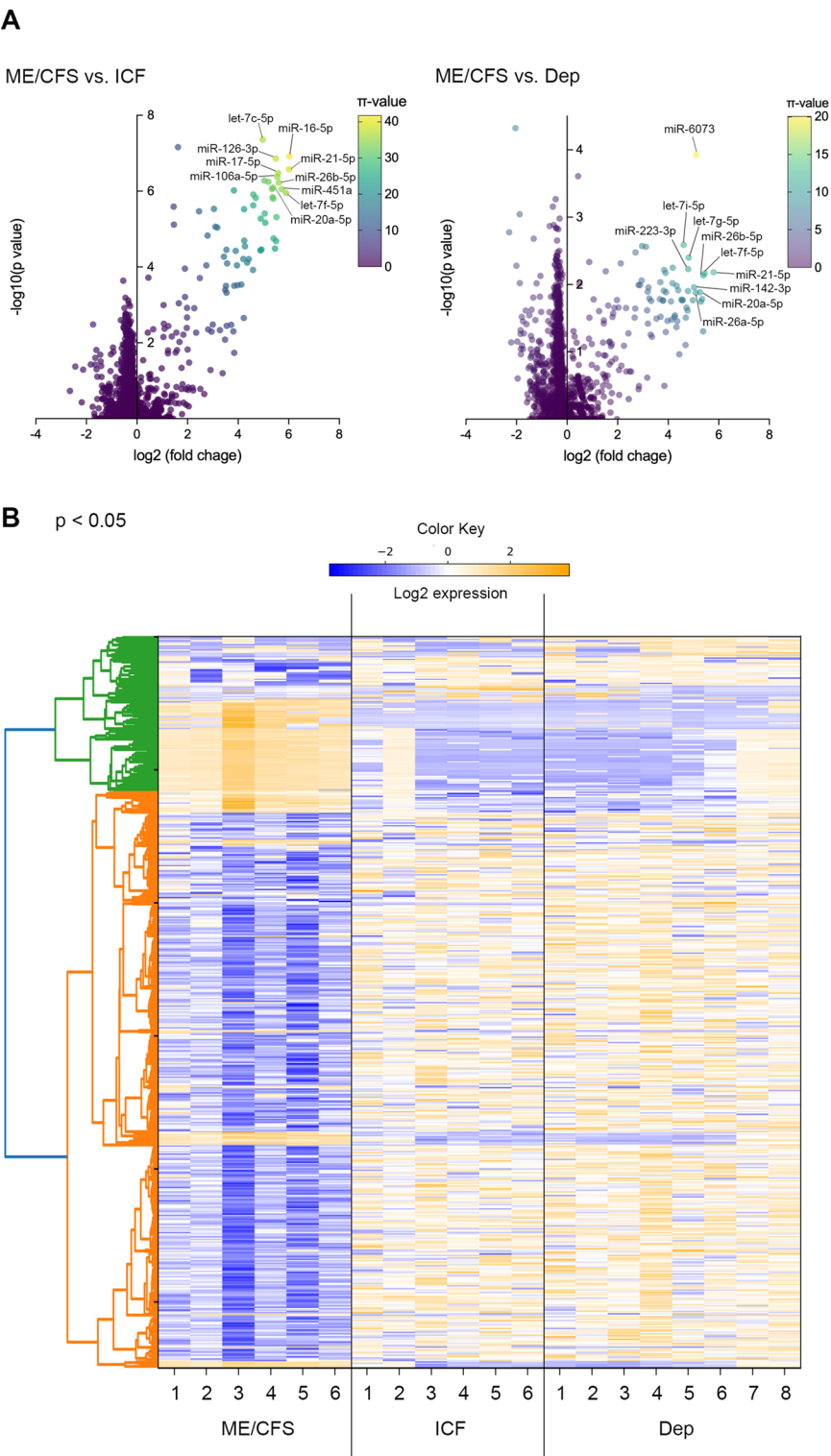
#### Identification of differentially expressed EV-miRNAs across the three groups

While the unique presence of certain miRNAs provides insight into the distinct pathophysiology of each disease, the development of clinical biomarkers requires a more comprehensive quantitative approach. Therefore, to establish a diagnostic framework, we evaluated the relative abundance of miRNAs across the groups. Entire EV-miRNA profiles were compared using volcano plots including  $\pi$ -values. (Fig. 3A and Supplementary Table 3). A significant number of up-regulated EV-miRNAs were identified in the ME/CFS group compared to the ICF or Dep groups. The top ten miRNAs by  $\pi$ -value in the ME/CFS group vs. ICF were miR-16-5p, miR-21-5p, miR-126-3p, let-7c-5p, miR-17-5p, miR-106a-5p, let-7f-5p, miR-26b-5p, miR-451a, and miR-20a-5p with large effect sizes (Supplementary Table 3). When compared to the Dep group, the top ten were miR-6073, miR-21-5p, let-7i-5p, let-7f-5p, let-7g-5p, miR-26b-5p, miR-223-3p, miR-20a-5p, miR-142-3p, and miR-26a-5p, also demonstrating large effect sizes (Supplementary Table 3). Notably, miR-21-5p, let-7f-5p, miR-26b-5p, and miR-20a-5p were common top-ranked miRNAs in both comparisons (Fig. 3A and Supplementary Table 3). A total of 550 EV-miRNAs were significantly differentially expressed in ME/CFS ( $p < 0.05$ ) (Fig. 3B and Supplementary Table 4), achieving  $87.9 \pm 4.8\%$  sensitivity and  $93.8 \pm 2.4\%$  specificity. Receiver operating characteristic analysis of prediction datasets generated by nine repeated three-fold cross-validation runs yielded mean AUC values of  $0.993 \pm 0.019$  for ME/CFS,  $0.903 \pm 0.143$  for ICF, and  $0.961 \pm 0.080$  for depression, indicating high classification performance across disease groups.

#### Identification of significant EV-miRNAs associated with the neuronal system in ME/CFS

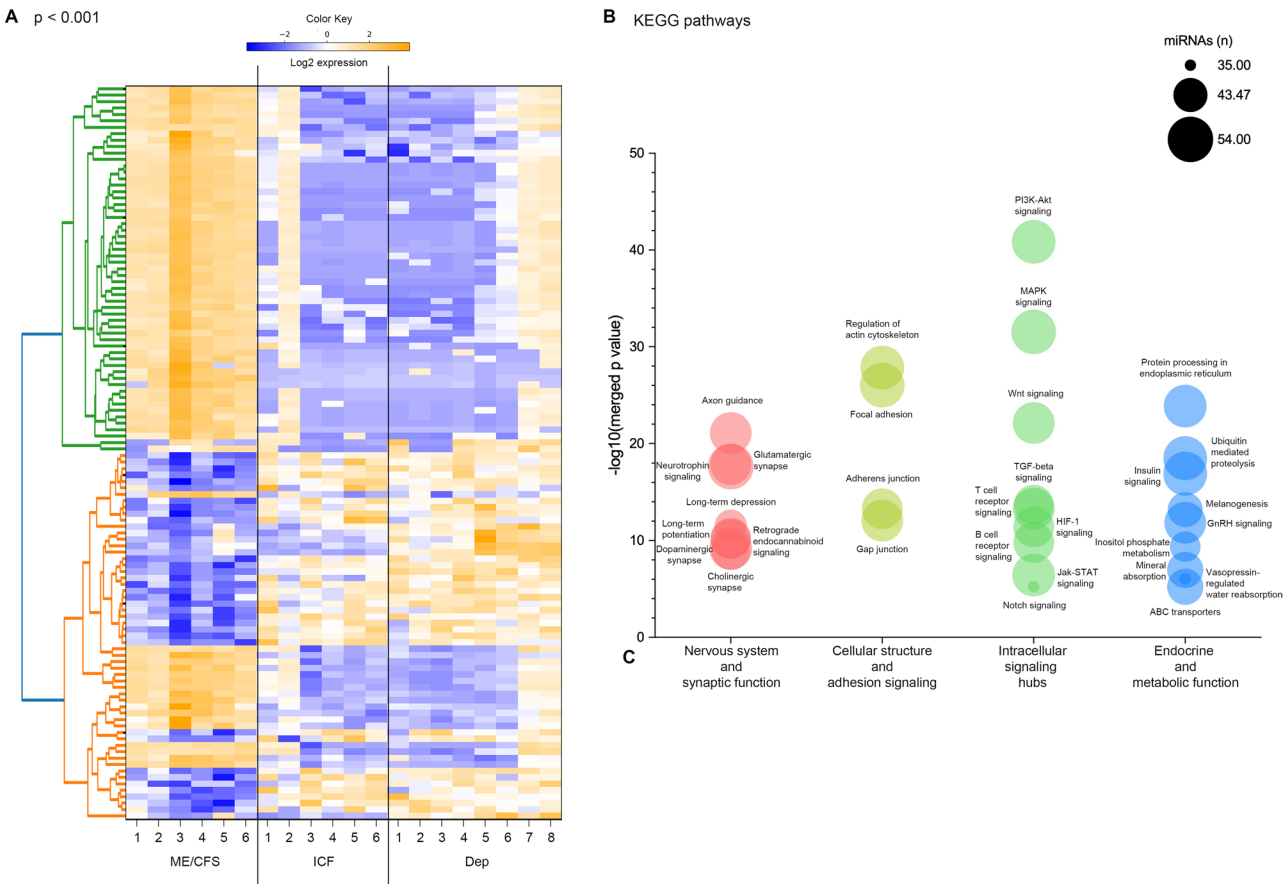
To further narrow down the candidate biomarkers, we applied a stricter statistical threshold ( $p < 0.001$ ) to the

EV-miRNA profiles. 114 EV-miRNAs remained significant (Fig. 4A and Supplementary Table 5), yielding a high diagnostic performance with  $87.0 \pm 4.8\%$  sensitivity and  $93.7 \pm 2.4\%$  specificity. The classifier achieved mean AUC values of  $1.000 \pm 0.000$ ,  $0.924 \pm 0.15$ , and  $0.96 \pm 0.07$  for ME/CFS, ICF, and depression, respectively, across nine repeated three-fold cross-validation runs (Supplementary Fig. 2). After adjusting for age, sex, and disease duration, the ANCOVA revealed that 23 out of 114 EV-miRNAs showed no significant difference between the groups (with F values, p values, and partial  $\eta^2$  values provided in Supplementary Table 6). Notably, the four common top-ranked EV-miRNAs shared across both comparisons (ME/CFS vs. ICF and Dep; Fig. 3A)—miR-21-5p, let-7f-5p, miR-26b-5p, and miR-20a-5p—were not among these 23 EV-miRNAs, maintaining their statistical significance. Of the remaining 91 significantly altered EV-miRNAs, 62 up-regulated EV-miRNAs in the ME/CFS group were significantly associated with 97 KEGG pathways (Supplementary Table 7). This stringent filtering identified 30 key KEGG pathways, which were categorized into four functional groups: (1) Nervous system and synaptic function, (2) Cellular structure and adhesion signaling, (3) Intracellular signaling hubs, and (4) Endocrine and metabolic functions (Fig. 4B). In the KEGG analysis, Axon guidance emerged as the most significantly enriched pathway, followed by glutamatergic synapse, neurotrophin signaling, long-term depression, long-term potentiation, retrograde endocannabinoid signaling, dopaminergic synapse, and cholinergic synapse (Fig. 4B), suggesting a dynamic modulation of neuronal architecture and synaptic plasticity in ME/CFS. Additionally, regulation of actin cytoskeleton, focal adhesion, adhesion junction, and gap junction were present as intracellular signaling hubs in KEGG pathway (Fig. 4B). Furthermore, KEGG analysis revealed several central regulatory cascades, including signaling pathways of PI3K-Akt, MAPK, Wnt, TGF- $\beta$ , T cell receptor, HIF-1, B cell receptor, Jak-STAT, and Notch (Fig. 4B). Focusing on systemic homeostasis, protein processing in endoplasmic reticulum, ubiquitin-mediated proteolysis, insulin signaling, melanogenesis, GnRH signaling, inositol phosphate metabolism, mineral absorption, vasopressin-regulated water reabsorption, and ABC transporters were predicted to regulate ME/CFS (Fig. 4B). These results suggest that the endocrine and metabolic dysfunction characteristic of ME/CFS is associated with specific alterations in protein trafficking and degradation pathways. These comprehensive miRNA profiles demonstrate that EV-miRNAs can identify ME/CFS with high sensitivity and specificity, reflecting the underlying neuronal dysfunction and systemic pathophysiology of the disease.



**Fig. 3** Differentially expressed EV-miRNAs in ME/CFS. **(A)** Volcano plots comparing miRNA expression between ME/CFS and ICF (left) and between ME/CFS and Dep (right). The  $\pi$ -value, a combined metric of fold change and statistical significance, was visualized by a color gradient, where higher  $\pi$ -values were indicated by yellow/purple colors. For each comparison, the top five miRNAs with the highest  $\pi$ -values were explicitly labeled and highlighted. **(B)** Heatmap analysis of 550 EV-miRNAs significantly differentially expressed among the three groups ( $p < 0.05$ ). ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome; ICF, idiopathic chronic fatigue; Dep, depression





**Fig. 4** Identification of significant ME/CFS-specific miRNA signatures and associated pathways. **(A)** Heatmap of 114 EV-miRNAs that remained significant with a stricter threshold ( $p < 0.001$ ). **(B)** Selected KEGG pathway analysis of 62 up-regulated miRNAs in the ME/CFS group. The analysis highlights significant enrichment in neuronal pathways. ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome; ICF, idiopathic chronic fatigue; Dep, depression

**Comparison of candidate EV-miRNA levels with a healthy control cohort**

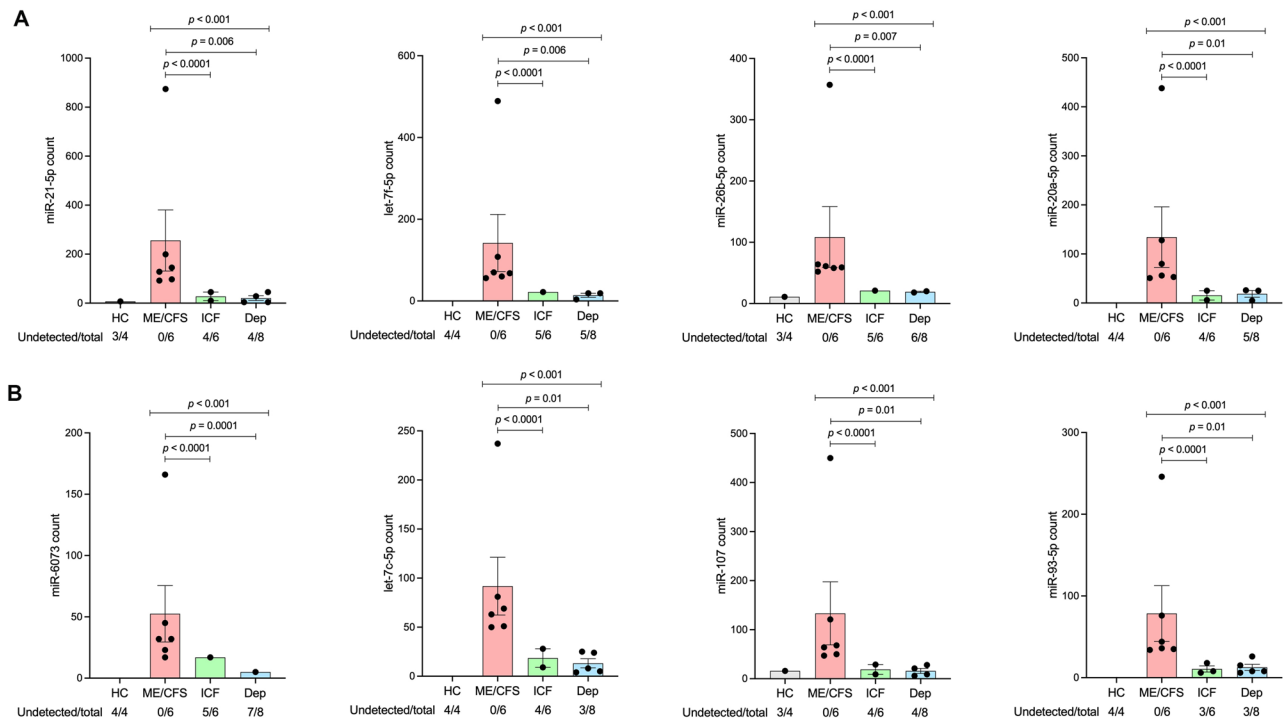
To preliminarily evaluate whether the elevation of these candidate EV-miRNA biomarkers is specific to ME/CFS, we examined their baseline EV-miRNA levels in an additional cohort of healthy controls (HC) (average age:  $56.00 \pm 3.46$  years; one female and three males). First, we examined the miRNA counts for the four common top-ranked EV-miRNAs identified in both volcano plot comparisons (Fig. 3A). These four EV-miRNAs (miR-21-5p, let-7f-5p, miR-26b-5p, and miR-20a-5p) demonstrated either undetectable or minimal counts in this HC group (Fig. 5A). Furthermore, we extended this verification to the top five EV-miRNAs derived from the stricter three-group comparison ( $p < 0.001$ ; Fig. 4A). Excluding miR-21-5p, which was already included in Fig. 5A, the profiles of the remaining four significantly altered EV-miRNAs (miR-6073, let-7c-5p, miR-107, and miR-93-5p) were plotted for the HC group (Fig. 5B). To ensure data transparency, the exact number of individuals with undetectable levels for each miRNA was explicitly indicated as “undetected / total” within the figures. In the HC group,

the levels of these target EV-miRNAs remained consistently low or below the limit of detection. Although the sample size was limited, these preliminary findings suggest that the marked elevation of these specific EV-miRNAs is not prominently observed in healthy controls, supporting their potential as specific diagnostic candidates for ME/CFS.

**Discussion**

Our study is the first to demonstrate the discriminatory potential between ME/CFS, ICF, and depression within this discovery cohort. A key finding was the identification of a unique subpopulation of EVs in ME/CFS patients characterized by high Calcein intensity and larger diameters (Fig. 1), suggesting that these EVs carry a dense molecular cargo reflecting chronic cellular stress.

Notably, let-7f-5p, along with miR-21-5p, miR-26b-5p, and miR-20a-5p, was identified as one of the most significantly up-regulated miRNAs in the circulating EVs of ME/CFS patients (Fig. 3A). These key candidate biomarkers maintained their strict statistical significance even after ANCOVA adjustment for age, sex, and disease



**Fig. 5** Comparison of candidate EV-miRNA biomarkers in a healthy control cohort. **(A)** Scatter plots showing the baseline EV-miRNA count for the four common top-ranked miRNAs identified in both two-group volcano plot comparisons (miR-21-5p, let-7f-5p, miR-26b-5p, and miR-20a-5p) across HC, ME/CFS, ICF, and Dep. **(B)** Scatter plots of the remaining four significant EV-miRNAs (miR-6073, let-7c-5p, miR-107, and miR-93-5p) derived from the strict three-group comparison ( $p < 0.001$ ), excluding miR-21-5p. HC, healthy control; ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome; ICF, idiopathic chronic fatigue; Dep, depression

duration. Furthermore, our preliminary evaluation with a small healthy control cohort demonstrated that the levels of these target EV-miRNAs remained consistently low or below the limit of detection (Fig. 5), supporting their potential relevance as specific diagnostic candidates for ME/CFS rather than universal markers of general fatigue or health status.

Among these candidates, the let-7 family is well known as a master regulator of metabolism, but recent evidence highlights its potent role as a signaling molecule in neuroinflammation. Specifically, let-7 family members can act as endogenous ligands for Toll-like receptor 7 (TLR7) [25], triggering pro-inflammatory cytokine release and neurodegeneration. Given the significant enrichment of let-7f-5p in ME/CFS-derived EVs, this pathway may drive the microglial-mediated neuroinflammation previously observed in PET imaging from our previous studies of ME/CFS patients [9].

This inflammatory state provides a direct molecular link to the disruption of the nervous system and synaptic function identified in KEGG analysis (Fig. 4B). The profound enrichment of pathways such as Axon guidance, glutamatergic synapse, neurotrophin signaling, and long-term potentiation suggested potential alterations in systemic neural “wiring” and synaptic plasticity. These neurological disruptions offer a compelling biological

explanation for the persistent “brain fog” and neurocognitive deterioration that serve as hallmarks of ME/CFS.

These neurological impairments are molecularly reinforced by the consistent dysregulation of cellular structure and intracellular signaling hubs. The convergence of our miRNA targets on regulation of the actin cytoskeleton and focal adhesion highlights a broad-scale modulation of cellular integrity and motility. Intriguingly, these findings at the miRNA level remarkably converge with our previous EV-proteomic analysis, which identified the actin, cytoskeleton, focal adhesion, and PI3K-Akt signaling as specific protein signatures of ME/CFS. These homeostatic failures appear to be driven by the alteration of universal signaling switches, specifically the PI3K-Akt pathway, MAPK cascade, and Wnt signaling pathway (Fig. 4B). By dampening these central hubs, EV-miRNAs might influence cellular states toward “metabolic hibernation” or inflexibility [5]. This hypothesis aligns with integrated pathophysiology models proposing that immune, metabolic, and oxidative stress pathways converge to drive the chronic exhaustion observed in patients [4, 5], which is further supported by recent evidence showing that the protein cargo of EVs undergoes significant changes after physical exertion in ME/CFS, uncovering disrupted energy metabolism linked to post-exertional malaise [32].

This state of systemic exhaustion is further reflected in the endocrine and metabolic functions identified in our analysis, such as insulin signaling, protein processing in the endoplasmic reticulum, and ABC transporters (Fig. 4B). Disruption in these pathways suggests a potential associated cascade that may affect the processing, trafficking, and secretion of critical hormones and metabolic factors, thereby potentially contributing to systemic endocrine instability and the profound fatigue reported by patients. As diagnostic and management strategies evolve toward digital multi-modal therapies, the identification of these specific EV-miRNA signatures offers a promising path toward the personalized medicine long called for in the field [4].

Several limitations should be considered when interpreting the present findings. First, the study cohort was relatively small, consisting of only 20 participants. Second, the classification model was evaluated using repeated cross-validation within the discovery cohort and was not validated in an independent external cohort. Therefore, the reported performance metrics should be interpreted as estimates of internal classification performance rather than real-world diagnostic accuracy. Third, although L1-regularized feature selection was applied to reduce model complexity and additional sensitivity analyses demonstrated consistent performance across different candidate miRNA sets, larger multicenter studies will be required to establish the robustness and generalizability of the identified EV-miRNA signatures. Fourth, while we adjusted for the major confounding factors, including sex, age, and disease duration, using ANCOVA, information regarding BMI and medication use was unavailable because these data were not included in the original study protocol. Furthermore, among the 114 miRNAs initially identified as significantly different between groups, 23 lost statistical significance after adjustment for these covariates and were excluded from subsequent KEGG pathway analyses. Since BMI and medication use may also influence miRNA expression, residual confounding cannot be completely excluded. Consequently, these results should be interpreted with caution, and future studies incorporating detailed assessments of BMI and medication use are needed to confirm and extend our findings. Ultimately, independent validation will be essential before clinical implementation can be considered. Fifth, because disease duration differed substantially across groups, the potential impact of long-term chronicity cannot be completely decoupled from disease specificity in this cohort. To reflect this with proper academic caution, we acknowledge that the identified EV-miRNA profiles may partially reflect the chronicity of the condition, and future studies incorporating duration-matched cohorts are needed to confirm our findings.

## Conclusions

Circulating EV-miRNAs in ME/CFS serve as integrative vehicles of disease, where the alteration of signaling hubs leads to a multi-layered failure of neuronal connectivity and metabolic homeostasis. These findings suggest that EVs are not merely objective biomarkers but active mediators of the disease's pathophysiology.

## Abbreviations

|        |                                                    |
|--------|----------------------------------------------------|
| ME/CFS | Myalgic encephalomyelitis/chronic fatigue syndrome |
| EVs    | Extracellular vesicles                             |
| ICF    | Idiopathic chronic fatigue                         |
| Dep    | Depression                                         |
| miRNA  | microRNA                                           |

## Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12967-026-08695-w>.

Supplementary Material 1: Supplementary Fig. 1. Circulating EV-miRNA profile and unique KEGG pathways in ICF and Dep patients. (A) Representative RNA electrophoresis profiles of EV-encapsulated RNAs. (B, C) KEGG pathway analysis of miRNAs exclusively detected in the ICF (B) and Dep (C) groups. Asterisks (★) denote pathways associated with neurological functions. ME/CFS, myalgic encephalomyelitis/chronic fatigue syndrome; ICF, idiopathic chronic fatigue; Dep, depression; EVs, extracellular vesicles; miRNA, microRNA.

Supplementary Material 2: Supplementary Fig. 2. Receiver operating characteristic (ROC) curves for repeated cross-validation prediction. Receiver operating characteristic (ROC) curves obtained from nine repeated three-fold cross-validation runs are shown for the one-versus-rest classification of (A) ME/CFS, (B) ICF, and (C) depression. Thin gray lines represent ROC curves from individual cross-validation runs ( $n=27$ ), whereas the colored line indicates the mean ROC curve for each class. Shaded regions represent  $\pm 1$  standard deviation of the true positive rate (TPR) across the repeated cross-validation runs. Mean area under the curve (AUC) values were  $1.000 \pm 0.000$  for ME/CFS,  $0.924 \pm 0.147$  for ICF, and  $0.957 \pm 0.071$  for depression. These results demonstrate the reproducibility of the classifier performance across repeated cross-validation while illustrating the variability arising from the limited sample size. (D) Aggregated confusion matrix summarizing prediction results from all 27 cross-validation runs. Values represent the percentage of predictions normalized within each true class (row normalization), illustrating the frequency with which samples from each class were assigned to the predicted classes across repeated cross-validation.

Supplementary Material 3: Supplementary Table 1: List of miRNAs uniquely detected in each group.

Supplementary Material 4: Supplementary Table 2: KEGG pathways regulated by miRNAs uniquely detected in each group.

Supplementary Material 5: Supplementary Table 3: Full list of EV-miRNAs for volcano plots with effect sizes.

Supplementary Material 6: Supplementary Table 4: Full list of significantly (adjusted  $p < 0.05$ ) altered EV-miRNAs used for heatmap analysis.

Supplementary Material 7: Supplementary Table 5: Full list of significantly (adjusted  $p < 0.001$ ) altered EV-miRNAs used for heat map analysis.

Supplementary Material 8: Supplementary Table 6: Intergroup comparison of EV-miRNAs adjusted for age, sex, and disease duration.

Supplementary Material 9: Supplementary Table 7: KEGG pathways targeted by the 62 significantly up-regulated miRNAs in ME/CFS.

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## Author contributions

Conceptualization: AE, SF. Data curation: SF, YN, RN, AE. Formal analysis: SF, TY. Funding acquisition: SF, AE. Supervision: HK, YN, YW. Writing – original draft: AE. Writing – review & editing: HK, YN, TY, YW, SF.

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## Data availability

All data generated or analyzed during this study are included in this published article and its supplementary information files.

## Declarations

### Ethics approval and consent to participate

The study was approved by the ethics committees of Kansai University Welfare of Health Sciences (Approval No. 09–06), Osaka City University Graduate School of Medicine (Approval No. 2151), and Mie University (Approval No. 1697 and H2018-069), and was conducted in accordance with the Declaration of Helsinki.

### Consent for publication

Not applicable.

### Competing interests

The authors declare that they have no competing interests.

### Author details

<sup>1</sup>Biobank Center, Mie University Hospital, 2-174 Edobashi, Tsu, Mie 514-8507, Japan

<sup>2</sup>Department of Gastroenterology and Hepatology, School of Medicine, Mie University, 2-174 Edobashi, Tsu, Mie 514-8507, Japan

<sup>3</sup>Division of Health Sciences, Osaka University Graduate School of Medicine, 1-7 Yamadaoka, Suita, Osaka 565-0871, Japan

<sup>4</sup>Fatigue & Sleep Clinic Yodoyabashi, Osaka 541-0043, Japan

<sup>5</sup>Center for Health Science Innovation, Osaka Metropolitan University, Osaka 558-8585, Japan

<sup>6</sup>Department of Biomolecular Engineering, Graduate School of Engineering, Nagoya University, Furo, Chikusa, Nagoya 464-8603, Japan

<sup>7</sup>Omiya City Clinic, Saitama 330-0854, Japan

<sup>8</sup>RIKEN Compass to Healthy Life Research Complex Program, Kobe 650-0047, Japan

<sup>9</sup>Kobe University Graduate School of Science, Technology and Innovation, Kobe 657-8501, Japan

<sup>10</sup>Department of Health Welfare Sciences, Kansai University of Welfare Sciences, Kashiwara 582-0026, Japan

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